Quant X

# A+ Chicago Futures Hot-Memory Universe

## Volatility Sensor Layer and Cross-Asset Market State Engine

Developer Prompt

Production research-infrastructure specification for Chicago futures market-state awareness, VIX/VVIX volatility sensing, point-in-time-safe feature generation, and backtest/live parity. This document is infrastructure guidance only and does not define a trading strategy.

Prepared as a developer handoff for Quant X.

Developer Task: A+ Chicago Futures Hot-Memory Universe, Volatility Sensor Layer, and Cross-Asset Market State Engine for Quant X

## Objective:

Build a production-grade market-state infrastructure layer for Quant X focused on Chicago futures trading, cross-asset context, and volatility-regime awareness.

This is not a trading strategy. Do not hard-code directional rules, alpha assumptions, entry logic, exit logic, or trade blockers. This layer exists to keep the correct futures universe, volatility sensors, event context, and mathematically defined state variables available to the research engine, backtest engine, replay engine, paper-trading layer, live engine, and Mission Control telemetry.

The implementation must be mathematically reproducible, point-in-time safe, feed-aware, latency-aware, and testable.

## Core Principle:

The system should not keep every futures contract hot. It should maintain:

1. HOT executable universe:

Always resident in memory. Live quote/trade/book state, rolling features, freshness status, and contract metadata are continuously updated.

2. HOT sensor universe:

Always resident as market-state sensors, but not necessarily tradable/order-book instruments. VIX and VVIX belong here.

3. WARM context universe:

Lower-frequency or lightweight state. Promoted to HOT when events, volatility, volume, spreads, news, or research requests justify it.

4. COLD reference universe:

Historical/reference only. Promoted only by explicit event trigger or research request.

Do not build a generic watchlist. Build a mathematically defined market-state layer.

## Critical Market-Structure Separation:

CME/CBOT/NYMEX/COMEX futures are Chicago futures instruments.

VIX and VVIX are Cboe volatility index sensors, not CME futures.

VX/VXM are tradable VIX futures on Cboe Futures Exchange if feed and broker support exist.

## Model them separately:

- VIX:

```
instrument_type = index_sensor
```

```
venue = Cboe index feed

tradable = false unless represented by derivatives elsewhere
```

- VVIX:

```
instrument_type = index_sensor

venue = Cboe index feed

tradable = false
```

- VX / VXM:

```
instrument_type = futures

venue = Cboe Futures Exchange

tradable = true only if feed and broker support exist
```

- ES, NQ, RTY, YM, ZT, ZF, ZN, ZB, UB, SR3, ZQ, CL, NG, GC, HG, 6E, etc.:

```
instrument_type = futures

venue = CME Group venue family as appropriate
```

Do not expect L2 book data for VIX or VVIX. They are index sensor streams. Missing, delayed, or unavailable VIX/VVIX data must degrade gracefully and must not break the CME futures engine.

# PHASE 1 - Instrument Registry and Symbol-Mapping Layer

Build or harden the instrument registry before building features.

## Each instrument must have:

- canonical_internal_symbol
- research_symbol
- broker_symbol
- data_vendor_symbol
- continuous_contract_symbol
- front_contract_symbol
- next_contract_symbol
- exchange
- venue
- asset_class
- instrument_type
- tradable flag
- tick_size

- tick_value
- contract_multiplier
- currency
- trading_hours
- session_calendar
- expiration_calendar
- roll_rule
- min_price_increment
- live_feed_status
- historical_feed_status
- order_book_available flag
- trade_print_available flag
- index_sensor_available flag
- point_in_time_safe flag
- data_delay_status: LIVE / DELAYED / HISTORICAL_ONLY / MISSING / DISABLED
- hot_memory_tier: HOT_EXECUTABLE / HOT_SENSOR / WARM / COLD
- failure_mode
- dependencies

Do not assume the historical symbol, live broker symbol, and execution symbol are the same. They are not always the same.

### Example:

ES research continuous series, ES front-month contract, ES broker execution symbol, ES data-vendor symbol, and ESM6/ESU6/ESZ6 individual contracts must be explicitly mapped.

Acceptance criteria for Phase 1:

- Every supported instrument has explicit symbol mapping.
- Missing feed access does not crash the system.
- Every instrument reports capability status.
- No instrument is treated as live-tradable unless broker and feed status confirm it.
- VIX/VVIX cannot be accidentally routed to execution.
- VX/VXM cannot be confused with VIX spot index values.

## PHASE 2 - Required HOT Universe

Implement the following HOT_EXECUTABLE futures universe.

## Equity Index Futures:

- ES / MES: S&P 500 futures
- NQ / MNQ: Nasdaq-100 futures
- RTY / M2K: Russell 2000 futures
- YM / MYM: Dow futures

## Purpose:

- U.S. equity beta
- tech/growth beta
- small-cap liquidity/growth stress
- industrial/value rotation
- cross-index divergence
- lead/lag propagation during macro and liquidity events

## Rates Futures:

- ZT: 2-Year Treasury futures
- ZF: 5-Year Treasury futures
- ZN: 10-Year Treasury futures
- ZB: 30-Year Treasury bond futures
- UB: Ultra Treasury bond futures
- SR3: 3-Month SOFR futures
- ZQ: Fed Funds futures

## Purpose:

- Fed path
- inflation repricing
- yield-curve pressure
- duration shock
- CPI/PPI/PCE/NFP/FOMC reaction mapping
- Treasury auction and Fed speaker context

## Energy Futures:

- CL / MCL: WTI crude oil
- NG: Henry Hub natural gas

## Purpose:

- inflation impulse
- geopolitical shock

- growth and demand stress
- energy-sector transmission
- EIA event context

## Metals Futures:

- GC / MGC: Gold
- HG: Copper

## Purpose:

- gold = real-rate, dollar, geopolitical, and liquidity-stress sensor
- copper = growth, industrial demand, China/global cycle sensor

## FX Futures:

- 6E: Euro FX futures

## Purpose:

- primary dollar-pressure proxy
- confirmation/contradiction layer for rates, gold, oil, and equities

HOT_SENSOR Volatility Layer:

- VIX Index
- VVIX Index
- VX front-month futures, if available
- VX second-month futures, if available
- VXM futures, if available
- optional: VIX9D, VIX3M, VXN, MOVE, SKEW, OVX, GVZ if feed/license supports them

## Minimum required hot set:

- ES
- NQ
- RTY
- YM
- ZT
- ZF
- ZN
- ZB
- UB
- SR3

- ZQ
- CL
- NG
- GC
- HG
- 6E
- VIX
- VVIX
- VX1, if available
- VX2, if available

### Important:

"Required" means supported in the registry and handled correctly. It does not mean the system must pretend a missing feed exists. Each item must report one of:

LIVE / DELAYED / HISTORICAL_ONLY / MISSING / DISABLED.

## PHASE 3 - WARM and COLD Universe

### WARM instruments:

- RB: RBOB gasoline
- HO: Heating oil
- SI: Silver
- 6J: Japanese yen
- 6B: British pound
- 6A: Australian dollar
- 6C: Canadian dollar
- ZC: Corn
- ZS: Soybeans
- ZW: Chicago wheat
- KE: Kansas City wheat
- ZL: Soybean oil
- ZM: Soybean meal

### COLD/event-triggered:

- LE: Live cattle
- GF: Feeder cattle
- HE: Lean hogs

- other commodities only if feed and research requirements justify them

### Promotion rules:

A WARM or COLD instrument may be promoted to HOT when one or more of these occur:

- macro calendar event relevance
- scheduled commodity report
- abnormal volume
- abnormal realized volatility
- abnormal spread widening
- abnormal price jump
- abnormal cross-asset divergence
- news/event detector relevance
- Research Director request
- replay/backtest experiment request
- risk layer context request

### Promotion must have:

- reason code
- timestamp
- triggering feature
- expected hot duration
- cooldown rule
- memory budget check
- demotion rule

No instrument should stay hot forever because of a stale event.

## PHASE 4 - Mathematical Feature Contract

Every feature must have a formal definition.

### For each feature, define:

- feature_name
- mathematical formula
- input fields
- sampling clock

- event time vs processing time
- lookback window
- update frequency
- normalization method
- missing-data policy
- outlier policy
- winsorization/clipping rule, if any
- session handling
- contract-roll handling
- point-in-time safety status
- latency stamp
- feature_version
- unit test reference

No feature should exist as vague code without a documented formula.

### Required microstructure features for tradable futures:

- mid_price
- spread_ticks
- spread_bps
- top_of_book_imbalance
- multi_level_book_imbalance, if depth exists
- microprice
- trade_sign_proxy
- signed_volume
- order_flow_imbalance
- realized_volatility over 1s, 5s, 15s, 30s, 1m, 5m
- return over 1s, 5s, 15s, 30s, 1m, 5m
- volume_zscore
- spread_zscore
- depth_zscore
- book_pressure
- sweep_flag
- quote_stuffing_or_burst_proxy
- stale_quote_flag
- crossed_or_locked_book_flag
- missing_depth_flag

- feed_latency_ms
- processing_latency_ms

### Required rates features:

- 2s10s proxy from ZT/ZN
- 5s30s proxy from ZF/ZB or ZF/UB
- 10s30s proxy from ZN/ZB or ZN/UB
- short_rate_path_proxy from SR3/ZQ
- rates_impulse_score
- curve_steepening_score
- curve_flattening_score
- equity_rates_divergence_score
- rates_led_move_flag
- equity_led_move_flag

### Required equity-index relative features:

- NQ_ES_relative_return
- RTY_ES_relative_return
- YM_ES_relative_return
- ES_NQ_spread_state
- index_dispersion_score
- growth_vs_broad_market_score
- smallcap_liquidity_stress_score
- equity_index_confirmation_score
- equity_index_divergence_score

### Required energy/metals/FX features:

- CL_equity_impulse_score
- CL_rates_impulse_score
- NG_energy_stress_score
- GC_rates_divergence_score
- GC_USD_proxy_divergence_score
- HG_growth_impulse_score
- 6E_USD_pressure_proxy
- USD_gold_confirmation_score
- USD_oil_confirmation_score

### Required VIX/VVIX/VX features:

- vix_level
- vvix_level
- vix_return_1m
- vix_return_5m
- vix_return_30m
- vix_return_1d
- vix_return_5d
- vvix_return_1d
- vvix_return_5d
- vvix_vix_ratio
- vx1_level, if available
- vx2_level, if available
- vx_front_basis = VX1 - VIX
- vx_term_spread = VX2 - VX1
- vx_curve_state = contango / flat / backwardation / unavailable
- vol_of_vol_pressure_score
- crash_convexity_pressure_score
- equity_vol_confirmation_score
- equity_vol_divergence_score
- quiet_index_stressed_vol_flag
- selloff_muted_vol_response_flag
- vix_sensor_freshness_status
- vvix_sensor_freshness_status

### Important:

Volatility features are state descriptors. They should not be hard-coded trade gates.

## PHASE 5 - Cross-Asset Market State Graph

Do not call this a causal graph unless formal causal testing is implemented.

### Name it:

Cross-Asset Market State Graph

The graph should represent dependency/context relationships, lead/lag observations, confirmation, contradiction, and regime state. It should not claim proven causality.

## Minimum graph relationships:

- rates complex -> equity index futures
- SR3/ZQ -> Fed path -> equity/rates regime
- VIX/VVIX/VX -> equity volatility regime
- 6E/USD proxy -> gold/oil/equity/rates interpretation
- CL/NG -> inflation and growth impulse
- GC -> real-rate/stress regime
- HG -> global growth/cyclical impulse
- NQ vs ES -> growth/tech/duration regime
- RTY vs ES -> liquidity/domestic growth regime
- VX term structure -> volatility carry/stress regime

## Graph outputs:

- dominant_driver
- dominant_driver_confidence
- secondary_driver
- regime_label
- regime_uncertainty_score
- cross_asset_confirmation_score
- cross_asset_contradiction_score
- stale_input_warning
- missing_sensor_warning
- volatility_regime
- rates_regime
- equity_dispersion_regime
- energy_shock_regime
- metals_stress_regime
- dollar_pressure_regime

## Critical rule:

This layer must not block trading. It emits state, confidence, contradiction, and warnings only. Trade/no-trade decisions belong to the model layer, risk layer, portfolio layer, or execution layer.

# PHASE 6 - Event Calendar and Event Windows

Integrate an event calendar layer with point-in-time timestamps.

## Required event categories:

- CPI
- PPI
- PCE
- NFP
- unemployment claims
- FOMC decision
- FOMC press conference
- Fed speakers
- Treasury auctions
- Treasury refunding
- EIA crude inventory
- EIA natural gas inventory
- USDA reports
- major geopolitical events
- exchange holidays
- contract roll dates
- futures expiration dates
- options expiration dates where relevant

## For each event:

- event_id
- event_type
- scheduled_timestamp
- actual_release_timestamp if available
- affected_asset_classes
- affected_symbols
- expected_importance
- source
- revision flag, if applicable
- point_in_time availability
- replay-safe flag

### Required event-window features:

- pre_event_volatility
- pre_event_spread
- pre_event_depth
- post_event_jump
- post_event_volatility
- first_mover_symbol
- rates_first_vs_equities_first
- vol_first_vs_equities_first
- cross_asset_confirmation_after_event
- cross_asset_contradiction_after_event

## PHASE 7 - Memory, Latency, and Degradation Rules

Define memory behavior explicitly.

### Default configurable hot-memory windows:

- tick/book/trade rolling buffer: configurable; default 30 to 120 minutes per HOT symbol
- 1-second bars: current session
- 1-minute bars: configurable; default 5 trading days
- 5-minute bars: configurable; default 30 trading days
- daily sensor history: configurable; default 2 years if available

### Latency/freshness:

### Each feed must report:

- exchange_timestamp
- receive_timestamp
- processing_timestamp
- feature_timestamp
- feed_latency_ms
- processing_latency_ms
- last_update_age_ms
- stale_status

### Default stale thresholds must be configurable by instrument type:

- futures L1/L2 feed stale threshold
- index sensor stale threshold

- delayed feed threshold
- historical-only status

### Under load:

- preserve HOT_EXECUTABLE instruments first
- preserve VIX/VVIX/VX sensor state second
- demote WARM instruments before HOT instruments
- never silently drop ES/NQ/rates state
- emit degradation telemetry

## PHASE 8 - Contract Roll and Historical Consistency

### Implement roll handling for futures:

- front contract
- next contract
- continuous contract
- volume-based roll
- open-interest-based roll if available
- calendar fallback
- raw contract series
- back-adjusted research series
- non-adjusted execution series

### Critical rule:

Do not mix adjusted research prices with live execution prices.

### Backtest/replay/live parity:

### The same feature definitions must work in:

- historical backtest
- market replay
- paper trading
- live trading

### Any feature that cannot be reconstructed historically must be marked:

```
live_only = true
```

and excluded from model training unless specifically approved.

# PHASE 9 - Feature Store and Audit Schema

### Persist generated features with:

- timestamp_event_time
- timestamp_receive_time
- timestamp_processing_time
- symbol
- contract
- continuous_contract_id
- feature_name
- feature_value
- lookback_window
- input_source
- calculation_version
- point_in_time_safe flag
- live_only flag
- latency_ms
- quality_flag
- stale_flag
- missing_input_flag
- roll_state
- event_context_id if applicable

Every model-facing feature must be traceable back to raw inputs and formula version.

# PHASE 10 - Mission Control Telemetry API

Do not build the UI first. Build a telemetry API first.

### Expose:

- HOT_EXECUTABLE instruments
- HOT_SENSOR instruments
- WARM instruments
- COLD instruments
- active promotions/demotions

- feed status by symbol
- stale feed warnings
- missing feed warnings
- latency by symbol
- VIX level and freshness
- VVIX level and freshness
- VX1/VX2 term structure if available
- volatility regime
- rates regime
- equity-index dispersion regime
- active dominant driver
- confirmation score
- contradiction score
- regime uncertainty score
- current roll state
- event-window state

Mission Control should consume this telemetry. It should not define backend logic.

## PHASE 11 - Testing and Validation

Add tests before marking complete.

### Required tests:

### Registry tests:

- every symbol has required metadata
- VIX/VVIX cannot route to execution
- VX/VXM are CFE futures, not CME futures
- missing feed status is handled cleanly
- broker/data/research symbols are not assumed identical

### Mathematical feature tests:

- formulas produce expected output on synthetic data
- normalization is stable
- missing data behavior is deterministic
- no divide-by-zero errors

- no lookahead leakage
- timestamps are point-in-time safe

### Roll tests:

- front contract selection works
- next contract selection works
- continuous contract construction works
- adjusted research series is separated from raw execution series

### Hot-memory tests:

- HOT instruments remain resident
- WARM promotion works
- demotion cooldown works
- under-load priority works
- stale feed detection works
- delayed index sensor does not crash futures layer

### Event-window tests:

- scheduled macro event attaches correctly
- event-window features reconstruct historically
- release timestamp handling is point-in-time safe
- post-event calculations do not leak into pre-event state

### Cross-asset state graph tests:

- graph emits state, not trade decisions
- missing VIX/VVIX degrades gracefully
- conflicting signals raise contradiction score
- stale inputs raise warnings
- no "do not trade" rule is enforced here

### Backtest/live parity tests:

- same feature code path works in backtest, replay, paper, and live
- live-only features are flagged
- model training excludes non-reconstructable features unless explicitly allowed

## PHASE 12 - Deliverables

Deliver the work in phases, not as one giant messy patch.

### Deliverable 1:

Instrument registry and symbol-mapping layer.

### Deliverable 2:

HOT/WARM/COLD memory manager.

### Deliverable 3:

VIX/VVIX/VX volatility sensor module.

### Deliverable 4:

Mathematical feature contract and initial feature set.

### Deliverable 5:

Cross-Asset Market State Graph.

### Deliverable 6:

Event calendar and event-window integration.

### Deliverable 7:

Contract-roll and backtest/live consistency hardening.

### Deliverable 8:

Telemetry API for Mission Control.

### Deliverable 9:

Test suite and documentation.

## Documentation must include:

- supported instruments
- unavailable instruments
- feed requirements
- broker requirements
- symbol mappings
- roll methodology
- feature definitions
- VIX/VVIX handling
- failure modes

- test coverage
- known limitations

## Acceptance Criteria:

1. The system has a clean registry for CME Group futures and Cboe volatility sensors.
2. VIX and VVIX are modeled as index sensors, not futures and not executable products.
3. VX/VXM are modeled as CFE futures only when feed and broker support exist.
4. Every instrument reports feed capability: LIVE / DELAYED / HISTORICAL_ONLY / MISSING / DISABLED.
5. Hot-memory behavior is explicit, configurable, and tested.
6. All model-facing features have documented formulas.
7. All features are point-in-time safe unless explicitly marked live_only.
8. Backtest, replay, paper, and live use the same feature definitions.
9. Adjusted research prices are never mixed with live execution prices.
10. Warm/cold instruments promote and demote based on explicit, logged triggers.
11. The Cross-Asset Market State Graph emits state and confidence only; it does not enforce trades.
12. Mission Control receives telemetry from the backend; it does not define backend logic.
13. The implementation adds situational awareness without constraining the model's search space.
14. No directional strategy, entry rule, exit rule, or trading bias is hard-coded.
15. Tests exist for registry, features, roll logic, VIX/VVIX separation, hot-memory promotion, stale feeds, event windows, and backtest/live parity.

## Final Instruction:

Implement this as institutional trading infrastructure. The goal is not to guess trades. The goal is to make Quant X mathematically aware of the correct Chicago futures universe, rates curve, equity-index complex, energy/metals/FX sensors, and volatility regime, including VIX, VVIX, and VX term structure, while preserving research integrity, point-in-time correctness, feed realism, and production reliability.

## Reference Dependencies

Use official exchange/vendor documentation for canonical contract specifications, feed definitions, and product classification. Do not infer symbol metadata manually.

- CME Group contract specifications and product metadata for CME/CBOT/NYMEX/COMEX futures.
- Cboe VIX product documentation for VIX Index, VIX futures, VIX term structure, and VVIX Index.
- Internal data catalog and broker/feed documentation for actual symbol mappings, entitlements, live/delayed status, historical availability, and execution support.